

Research Artifacts Repository

This section provides a central reference for the key scripts, data files, configuration files, and reports generated or utilized during this research. Generative AI tools provided assistance during the design and debugging phases for many of the scripts listed below. The complete collection of artifacts corresponding to this thesis has been archived on Zenodo and is available via the persistent DOI: <https://doi.org/10.5281/zenodo.15271232>. The table below indexes key items within this collection and provides links to browse specific files and directories in the associated public GitHub repository.

Table 1: Index of Research Artifacts

ID	Description	Type	Primary Section(s)	External Link
S1	Scripts for fetching PR data from GitHub API	Script	5.1.1, 6.1	View File
S2	Scripts for auto-mated PR scoring using zero-shot classification	Script	5.1.1, 6.1	View File
S3	Scripts for identifying top-scoring PRs per competency	Script	5.1.1	View File
S4	Script for PR quality analysis (readability, sentiment, counts)	Script	6.1, 6.3	View File
S5	Script for comparing PR versions (similarity metrics)	Script	6.3	View File
S6	Script orchestrating model agnosticism analysis	Script	6.3	View File
S7	Script for processing and analyzing survey results	Script	6.2, 5.5	View File

Continued on next page

Table 1 – continued from previous page

ID	Description	Type	Primary Section(s)	External Link
S8	Script for holistic PR analysis (EDA, repo comparison, PCA, clustering)	Script	6.1	View File
S9	Survey load balancer script (JS/HTML)	Script	5.3.5	View File
D1	Configuration for Latin Square design	Config	5.3.4	View File
D2	Dataset of preprocessed and scored PRs	Data (Processed)	5.1.1, 6.1	View File
D3	Aggregated repository statistics (scores, counts)	Data (Processed)	6.1	View File
D4	Repository correlation and clustering analysis data	Data (Processed)	6.1	View File
D5	Combined processed survey responses	Data (Processed)	6.2	Not publicly available due to privacy commitments.
D6	Qualitative remarks extracted from survey	Data (Processed)	6.2.5	Not publicly available due to privacy commitments.
D7	Raw model agnosticism quality/similarity comparisons	Data (Raw)	6.3	View File
D8	Summarized data analysis results	Data (Processed)	6.3	View Directory
D9	Generated survey analysis reports and tables	Report/Data (Processed)	6.2	View Directory
Continued on next page				

Table 1 – continued from previous page

ID	Description	Type	Primary Section(s)	External Link
D10	Model agnosticism statistical test report text	Report	6.3	View File
D11	Raw PR quality analysis results	Data (Raw)	5.4.1, 6.2.5	View File
D12	Interactive clustering visualization	Data (Interactive)	6.1	View File
P1	Prompt templates for generating PR variations	Prompt	5.2.1	View Directory
V1	Additional Detailed Visualizations	Visualization	6.2.3	View Directory

Repositories Considered for Initial Fetching

The following list details the GitHub repositories initially identified and considered for fetching Pull Requests during the automated analysis phase (Section 6.1, Section 5.1.1). This broad selection aimed to cover diverse domains within software engineering. Note that filtering criteria (e.g., comment count, date range) and subsequent expert validation (Section 5.1.2) meant that the final 6 PRs used as stimuli in the human evaluation survey originated from a subset of these or potentially related repositories, and not all repositories listed here necessarily contributed usable data to the final analysis stages after filtering.

Table 2: List of GitHub Repositories Initially Considered for PR Fetching

Category	Owner	Repository
Automation & DevOps	ansible	ansible
Automation & DevOps	grafana	grafana
Automation & DevOps	jenkinsci	jenkins
Automation & DevOps	puppetlabs	puppet

Continued on next page

Table 2 – continued from previous page

Category	Owner	Repository
Automation & DevOps	actions	runner
Automation & DevOps	saltstack	salt
Backend & Cloud Infrastructure	apache	cassandra
Backend & Cloud Infrastructure	containerd	containerd
Backend & Cloud Infrastructure	docker	cli
Backend & Cloud Infrastructure	elastic	elasticsearch
Backend & Cloud Infrastructure	apache	hadoop
Backend & Cloud Infrastructure	hashicorp	terraform
Backend & Cloud Infrastructure	istio	istio
Backend & Cloud Infrastructure	apache	kafka
Backend & Cloud Infrastructure	moby	moby
Backend & Cloud Infrastructure	nodejs	node
Backend & Cloud Infrastructure	pandas-dev	pandas
Backend & Cloud Infrastructure	prometheus	prometheus
Databases & Storage	cockroachdb	cockroach
Databases & Storage	etcd-io	etcd
Databases & Storage	influxdata	influxdb
Databases & Storage	mariadb	server
Databases & Storage	mongodb	mongo
Databases & Storage	postgres	postgres
Databases & Storage	redis	redis
Databases & Storage	facebook	rocksdb
Development Tools	bazelbuild	bazel
Development Tools	eslint	eslint
Development Tools	git	git
Development Tools	neovim	neovim
Development Tools	openresty	openresty
Development Tools	prettier	prettier
Game Engines & Graphics	blender	blender
Game Engines & Graphics	godotengine	godot
Game Engines & Graphics	openframeworks	openFrameworks
Game Engines & Graphics	pygame	pygame

Continued on next page

Table 2 – continued from previous page

Category	Owner	Repository
Game Engines & Graphics	defold	defold
Machine Learning & AI	facebookresearch	detectron2
Machine Learning & AI	fastai	fastai
Machine Learning & AI	scikit-learn	scikit-learn
Machine Learning & AI	tensorflow	tensorflow
Machine Learning & AI	huggingface	transformers
Machine Learning & AI	pytorch	pytorch
Networking	cilium	cilium
Networking	coredns	coredns
Networking	zephyrproject-rtos	zephyr
Operating Systems & Low-Level Software	freebsd	freebsd
Operating Systems & Low-Level Software	openbsd	src
Operating Systems & Low-Level Software	qemu	qemu
Operating Systems & Low-Level Software	reactos	reactos
Operating Systems & Low-Level Software	systemd	systemd
Operating Systems & Low-Level Software	xen-project	xen
Programming Languages	openjdk	jdk
Programming Languages	python	cpython
Programming Languages	golang	go
Programming Languages	llvm	llvm-project
Programming Languages	rust-lang	rust
Programming Languages	apple	swift
Programming Languages	microsoft	TypeScript
Security & Cryptography	letsencrypt	boulder
Security & Cryptography	sigstore	cosign
Security & Cryptography	libressl	libressl
Security & Cryptography	openssl	openssl
Security & Cryptography	hashicorp	vault
Web & API Frameworks	aspnet	aspnetcore
Web & API Frameworks	django	django
Web & API Frameworks	expressjs	express
Web & API Frameworks	fastify	fastify

Continued on next page

Table 2 – continued from previous page

Category	Owner	Repository
Web & API Frameworks	pallets	flask
Web & API Frameworks	rails	rails
Web & API Frameworks	spring-projects	spring-framework
Web & Frontend Frameworks	angular	angular
Web & Frontend Frameworks	emberjs	ember.js
Web & Frontend Frameworks	flutter	flutter
Web & Frontend Frameworks	vercel	next.js
Web & Frontend Frameworks	facebook	react
Web & Frontend Frameworks	solidjs	solid
Web & Frontend Frameworks	sveltejs	svelte
Web & Frontend Frameworks	tailwindlabs	tailwindcss
Web & Frontend Frameworks	vuejs	vue

Competency Definitions

Architecture Impact: Evaluates architectural implications and design pattern changes.

Code Quality: Assesses code correctness, error handling, and technical debt.

Collaboration: Evaluates communication clarity and team collaboration aspects.

Maintainability: Analyzes documentation, modularity, and long-term viability.

User Impact: Assesses user-facing changes and UX considerations.

Technical Leadership: Measures influence on technical direction and mentoring.

High-Level Prototype Workflow Diagram

The workflow operates as follows:

1. A developer closes a Pull Request.
2. The GitHub Action is triggered by the 'pull_request.closed' event.

3. The Action fetches relevant PR metadata (e.g., changed files, commit messages, comments, contributor identities).
4. It identifies the competencies pre-selected by the contributors for feedback.
5. The PR data and selected competencies are dynamically structured into a prompt template.
6. The tailored prompt is sent to a configured LLM API (e.g., OpenAI, Gemini).
7. The LLM response, containing suggested feedback, is received.
8. The Action processes the response, potentially parsing structured output or extracting relevant feedback snippets.
9. Personalized feedback messages (nudges) are constructed and delivered to the relevant contributors (e.g., via GitHub comments, issues, or another notification channel), linking the feedback to their specific contributions and selected competencies.